

ME101: Engineering Mechanics (3 1 0 8)

2023-2024 (II Semester)

Division II & IV

Instructor

Hemant B. Kaushik

Room M-003

Department of Civil Engineering

IIT Guwahati

hemantbk@iitg.ac.in

Phone: 258 2427

Web: www.iitg.ac.in/hemantbk

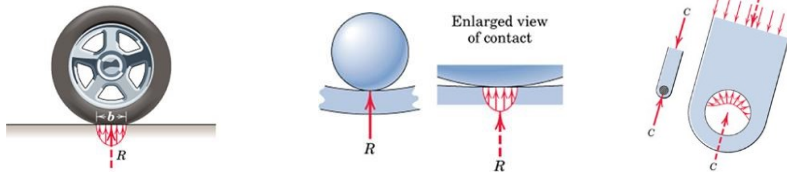
ME101 - Division II & IV

Hemant B Kaushik

¹ L09

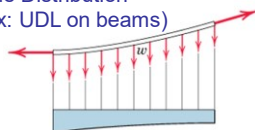
Center of Mass and Centroids

Concentrated Forces: If dimension of the contact area is negligible compared to other dimensions of the body → the contact forces may be treated as Concentrated Forces



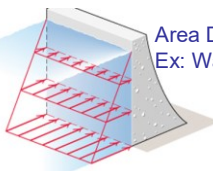
Distributed Forces: If forces are applied over a region whose dimension is not negligible compared with other pertinent dimensions → proper distribution of contact forces must be accounted for to know intensity of force at any location.

Line Distribution
(Ex: UDL on beams)

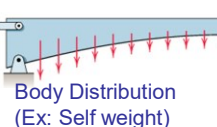


ME101 - Division II & IV

Area Distribution
(Ex: Water Pressure)



Hemant B Kaushik



Body Distribution
(Ex: Self weight)

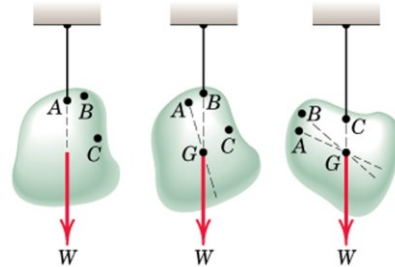
2

Center of Mass and Centroids

Center of Mass

A body of mass m in equilibrium under the action of tension in the cord, and resultant W of the gravitational forces acting on all particles of the body.

- The resultant is collinear with the cord



Suspend the body from different points on the body

- Dotted lines show lines of action of the resultant force in each case.
- These lines of action will be concurrent at a single point G

As long as dimensions of the body are smaller compared with those of the earth.

- we assume uniform and parallel force field due to the gravitational attraction of the earth.

The unique **Point G** is called the Center of Gravity of the body (CG)

ME101 - Division II & IV

Hemant B Kaushik

3

Center of Mass and Centroids

Determination of CG

- Apply Principle of Moments

Moment of resultant gravitational force W about any axis equals sum of the moments about the same axis of the gravitational forces dW acting on all particles treated as infinitesimal elements.

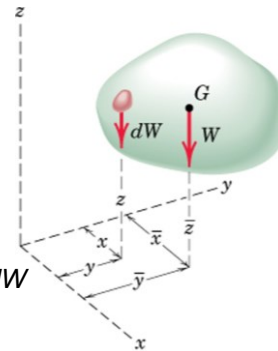
Weight of the body $W = \int dW$

Moment of weight of an element (dW) @ x-axis = $y dW$

Sum of moments for all elements of body = $\int y dW$

From Principle of Moments: $\int y dW = \bar{y} W$

$$\bar{x} = \frac{\int x dW}{W} \quad \bar{y} = \frac{\int y dW}{W} \quad \bar{z} = \frac{\int z dW}{W}$$



→ Numerator of these expressions represents the **sum of the moments**;
Product of W and corresponding coordinate of G represents
the **moment of the sum** → Moment Principle.

ME101 - Division II & IV

Hemant B Kaushik

4

Center of Mass and Centroids

Determination of CG

$$\bar{x} = \frac{\int x dW}{W} \quad \bar{y} = \frac{\int y dW}{W} \quad \bar{z} = \frac{\int z dW}{W}$$

Substituting $W = mg$ and $dW = gdm$

$$\rightarrow \bar{x} = \frac{\int x dm}{m} \quad \bar{y} = \frac{\int y dm}{m} \quad \bar{z} = \frac{\int z dm}{m}$$

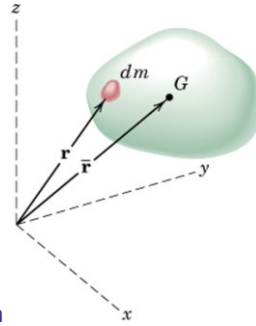
In vector notations:

Position vector for elemental mass: $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

Position vector for mass center G: $\bar{\mathbf{r}} = \bar{x}\mathbf{i} + \bar{y}\mathbf{j} + \bar{z}\mathbf{k}$

$$\rightarrow \bar{\mathbf{r}} = \frac{\int \mathbf{r} dm}{m}$$

The above equations are the components of this single vector equation



Density ρ of a body = **mass per unit volume**

$\rightarrow$ Mass of a differential element of volume $dV \rightarrow dm = \rho dV$

$\rightarrow \rho$ may not be constant throughout the body

$$\bar{x} = \frac{\int x \rho dV}{\int \rho dV} \quad \bar{y} = \frac{\int y \rho dV}{\int \rho dV} \quad \bar{z} = \frac{\int z \rho dV}{\int \rho dV}$$

ME101 - Division II & IV

Hemant B Kaushik

5

Center of Mass and Centroids

Center of Mass: Following equations independent of g

$$\bar{x} = \frac{\int x dm}{m} \quad \bar{y} = \frac{\int y dm}{m} \quad \bar{z} = \frac{\int z dm}{m} \quad \bar{\mathbf{r}} = \frac{\int \mathbf{r} dm}{m} \quad \bar{x} = \frac{\int x \rho dV}{\int \rho dV} \quad \bar{y} = \frac{\int y \rho dV}{\int \rho dV} \quad \bar{z} = \frac{\int z \rho dV}{\int \rho dV}$$

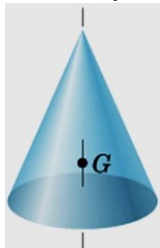
$\rightarrow$ They define a unique point, which is a function of distribution of mass

$\rightarrow$ This point is **Center of Mass (CM)**

$\rightarrow$ **CM coincides with CG as long as gravity field is treated as uniform and parallel**

$\rightarrow$ CG or CM may lie outside the body

CM always lie on a line or a plane of symmetry in a homogeneous body



Right Circular Cone

CM on central axis

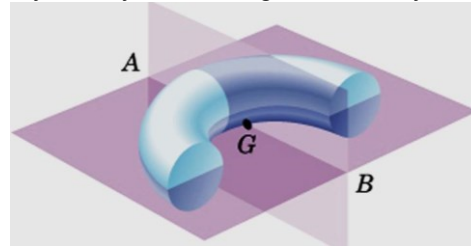
ME101 - Division II & IV



Half Right Circular Cone

CM on vertical plane of symmetry

Hemant B Kaushik



Half Ring

CM on intersection of two planes of symmetry (line AB)

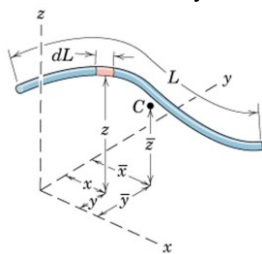
6

Center of Mass and Centroids

Centroids of Lines, Areas, and Volumes

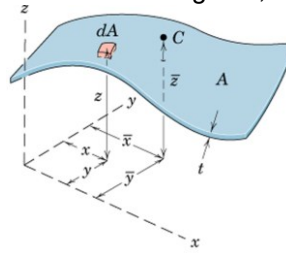
Centroid is a geometrical property of a body

→ When density of a body is uniform throughout, centroid and CM coincide



Lines: Slender rod, Wire
Cross-sectional area = A
 ρ and A are constant over L
 $dm = \rho A dL$; Centroid = CM

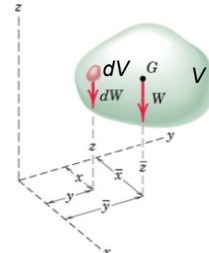
$$\bar{x} = \frac{\int x dL}{L} \quad \bar{y} = \frac{\int y dL}{L} \quad \bar{z} = \frac{\int z dL}{L}$$



Areas: Body with small but constant thickness t
Cross-sectional area = A
 ρ and A are constant over A
 $dm = \rho t dA$; Centroid = CM

$$\bar{x} = \frac{\int x dA}{A} \quad \bar{y} = \frac{\int y dA}{A} \quad \bar{z} = \frac{\int z dA}{A}$$

Numerator = First moments of Area
Hemant B Kaushik



Volumes: Body with volume V
 ρ constant over V
 $dm = \rho dV$
Centroid = CM

$$\bar{x} = \frac{\int x dV}{V} \quad \bar{y} = \frac{\int y dV}{V} \quad \bar{z} = \frac{\int z dV}{V}$$

ME101 - Division II & IV

7

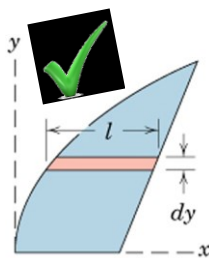
Center of Mass and Centroids

Centroids of Lines, Areas, and Volumes

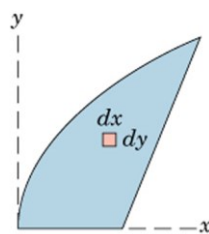
Guidelines for Choice of Elements for Integration

- **Order of Element Selected for Integration**

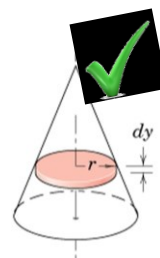
A first order differential element should be selected in preference to a higher order element → only one integration should cover the entire figure



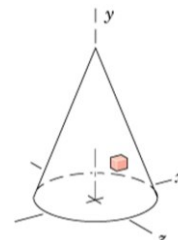
$$A = \int dA = \int l dy$$



$$A = \iint dx dy$$



$$V = \int dV = \int \pi r^2 dy$$



$$V = \iiint dx dy dz$$

ME101 - Division II & IV

Hemant B Kaushik

8

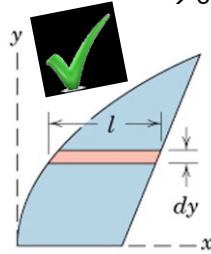
Center of Mass and Centroids

Centroids of Lines, Areas, and Volumes

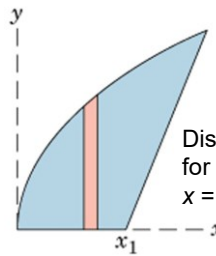
Guidelines for Choice of Elements for Integration

- **Continuity**

Choose an element that can be integrated in one continuous operation to cover the entire figure → the function representing the body should be continuous
→ only one integral will cover the entire figure



Continuity in the expression for the width of the strip



Discontinuity in the expression for the height of the strip at $x = x_1$

ME101 - Division II & IV

Hemant B Kaushik

9

Center of Mass and Centroids

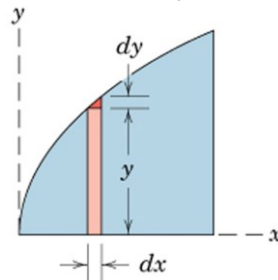
Centroids of Lines, Areas, and Volumes

Guidelines for Choice of Elements for Integration

- **Discarding Higher Order Terms**

Higher order terms may always be dropped compared with lower order terms
→ there is no error in the limit

Vertical strip of area under the curve is given by the first order term → $dA = ydx$
The second order triangular area $0.5dxdy$ may be discarded



ME101 - Division II & IV

Hemant B Kaushik

10

Center of Mass and Centroids

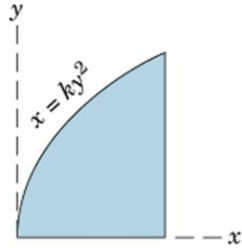
Centroids of Lines, Areas, and Volumes

Guidelines for Choice of Elements for Integration

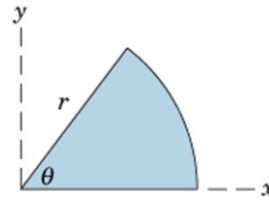
- **Choice of Coordinates**

Coordinate system should best match the boundaries of the figure

→ easiest coordinate system that satisfies boundary conditions should be chosen



Boundaries of this area (not circular) can be easily described in rectangular coordinates



Boundaries of this circular sector are best suited to polar coordinates

ME101 - Division II & IV

Hemant B Kaushik

11

Center of Mass and Centroids

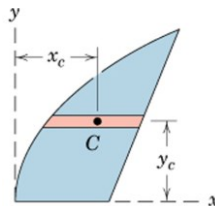
Centroids of Lines, Areas, and Volumes

Guidelines for Choice of Elements for Integration

- **Centroidal Coordinate of Differential Elements**

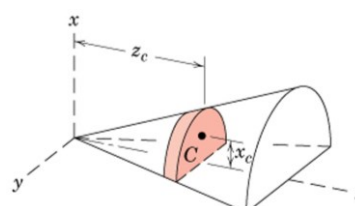
While expressing moment of differential elements, take coordinates of the centroid of the differential element as lever arm

(not the coordinate describing the boundary of the area)



Modified Equations

$$\bar{x} = \frac{\int x_c dA}{A} \quad \bar{y} = \frac{\int y_c dA}{A} \quad \bar{z} = \frac{\int z_c dA}{A}$$



$$\bar{x} = \frac{\int x_c dV}{V} \quad \bar{y} = \frac{\int y_c dV}{V} \quad \bar{z} = \frac{\int z_c dV}{V}$$

ME101 - Division II & IV

Hemant B Kaushik

12

Center of Mass and Centroids

Centroids of Lines, Areas, and Volumes

Guidelines for Choice of Elements for Integration

1. Order of Element Selected for Integration
2. Continuity
3. Discarding Higher Order Terms
4. Choice of Coordinates
5. Centroidal Coordinate of Differential Elements

$$\bar{x} = \frac{\int x dL}{L} \quad \bar{y} = \frac{\int y dL}{L} \quad \bar{z} = \frac{\int z dL}{L}$$

$$\bar{x} = \frac{\int x_c dA}{A} \quad \bar{y} = \frac{\int y_c dA}{A} \quad \bar{z} = \frac{\int z_c dA}{A}$$

$$\bar{x} = \frac{\int x_c dV}{V} \quad \bar{y} = \frac{\int y_c dV}{V} \quad \bar{z} = \frac{\int z_c dV}{V}$$

ME101 - Division II & IV

Hemant B Kaushik

13

Center of Mass and Centroids

Examples: Centroids

Locate the centroid of the circular arc

Solution: Polar coordinate system is better

Since the figure is symmetric: centroid lies on the x axis

Differential element of arc has length $dL = r d\theta$

Total length of arc: $L = 2\alpha r$

x-coordinate of the centroid of differential element: $x = r \cos \theta$

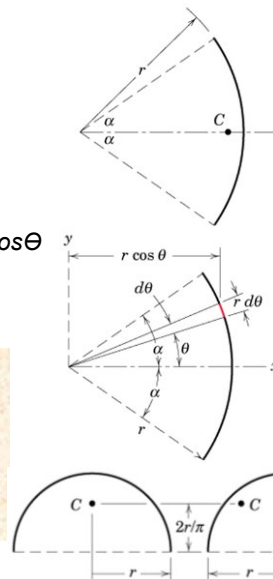
$$\bar{x} = \frac{\int x dL}{L} \quad \bar{y} = \frac{\int y dL}{L} \quad \bar{z} = \frac{\int z dL}{L}$$

$$[L\bar{x} = \int x dL] \quad (2\alpha r)\bar{x} = \int_{-\alpha}^{\alpha} (r \cos \theta) r d\theta$$

$$2\alpha r\bar{x} = 2r^2 \sin \alpha$$

$$\bar{x} = \frac{r \sin \alpha}{\alpha}$$

For a semi-circular Arc: $2\alpha = \pi \rightarrow$ centroid lies at $2r/\pi$



ME101 - Division II & IV

Hemant B Kaushik

14

Center of Mass and Centroids

Examples: Centroids

Locate the centroid of the triangle along h from the base

Solution:

$$dA = xdy ; x/(h-y) = b/h$$

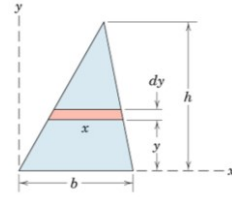
$$\text{Total Area, } A = \frac{1}{2}(bh)$$

$$\bar{x} = \frac{\int x_c dA}{A} \quad \bar{y} = \frac{\int y_c dA}{A} \quad \bar{z} = \frac{\int z_c dA}{A}$$

$$y_c = y$$

$$[A\bar{y} = \int y_c dA] \quad \frac{bh}{2} \bar{y} = \int_0^h y \frac{b(h-y)}{h} dy = \frac{bh^2}{6}$$

$$\text{and} \quad \bar{y} = \frac{h}{3}$$



ME101 - Division II & IV

Hemant B Kaushik

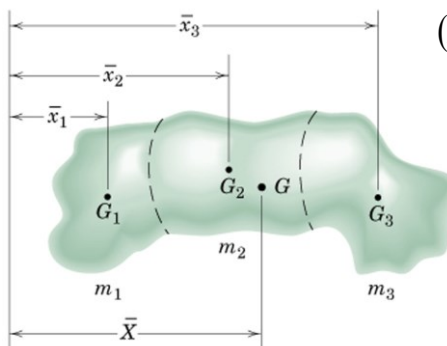
15

Center of Mass and Centroids

Composite Bodies and Figures

Divide bodies or figures into several parts such that their mass centers can be conveniently determined

→ Use Principle of Moment for all finite elements of the body



x-coordinate of the center of mass of the whole

$$(m_1 + m_2 + m_3)\bar{X} = m_1\bar{x}_1 + m_2\bar{x}_2 + m_3\bar{x}_3$$

Mass Center Coordinates can be written as:

$$\bar{X} = \frac{\sum m\bar{x}}{\sum m} \quad \bar{Y} = \frac{\sum m\bar{y}}{\sum m} \quad \bar{Z} = \frac{\sum m\bar{z}}{\sum m}$$

m's can be replaced by L's, A's, and V's for lines, areas, and volumes

ME101 - Division II & IV

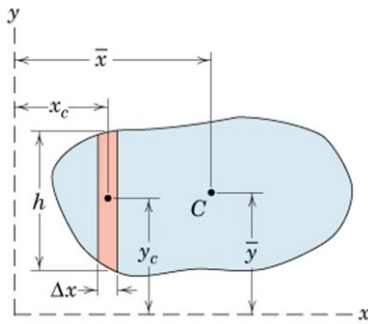
Hemant B Kaushik

16

Center of Mass and Centroids: Composite Bodies and Figures

Integration vs Appx Summation: Irregular Area

In some cases, the boundaries of an area or volume might not be expressible mathematically or in terms of simple geometrical shapes → Appx Summation may be used instead of integration



Divide the area into several strips
 Area of each strip = $h\Delta x$
 Moment of this area about x- and y-axis
 = $(h\Delta x)y_c$ and $(h\Delta x)x_c$
 → Sum of moments for all strips
 divided by the total area will give
 corresponding coordinate of the centroid

$$\bar{x} = \frac{\sum Ax_c}{\sum A} \quad \bar{y} = \frac{\sum Ay_c}{\sum A}$$

Accuracy may be improved by reducing the width of the strip

ME101 - Division II & IV

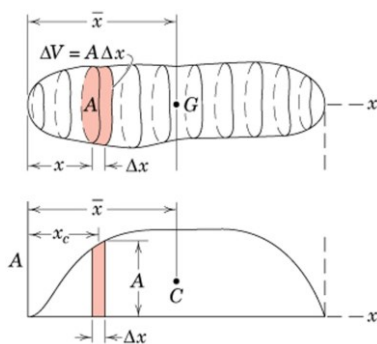
Hemant B Kaushik

17

Center of Mass and Centroids: Composite Bodies and Figures

Integration vs Appx Summation: Irregular Volume

Reduce the problem to one of locating the centroid of area
 → Appx Summation may be used instead of integration



Divide the area into several strips
 Volume of each strip = $A\Delta x$
 Plot all such A against x.
 → Area under the plotted curve
 represents volume of whole body and the
 x-coordinate of the centroid of the area
 under the curve is given by:

$$\bar{x} = \frac{\sum (A\Delta x)x_c}{\sum A\Delta x} \Rightarrow \bar{x} = \frac{\sum Vx_c}{\sum V}$$

Accuracy may be improved by reducing the width of the strip

ME101 - Division II & IV

Hemant B Kaushik

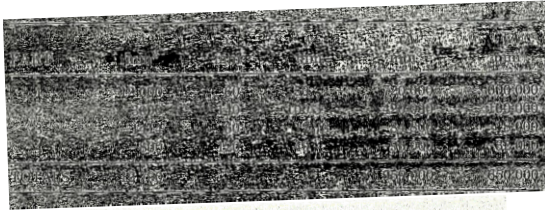
18

Center of Mass and Centroids: Composite Bodies and Figures

Example:

Locate the centroid of the shaded area

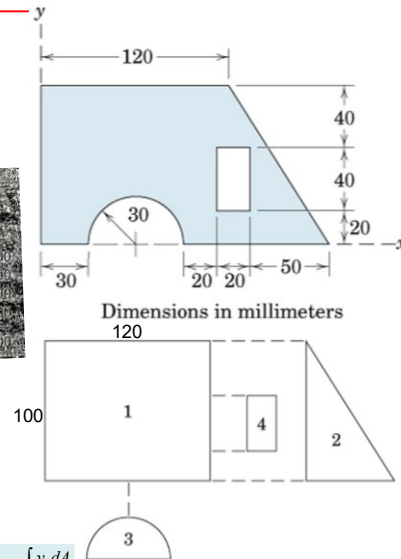
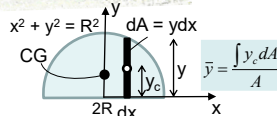
Solution: Divide the area into four elementary shapes: Total Area = $A_1 + A_2 - A_3 - A_4$



$$\begin{aligned} \bar{X} &= \frac{\sum A\bar{x}}{\sum A} & \bar{X} &= \frac{959\,000}{12\,790} = 75.0 \text{ mm} \\ \bar{Y} &= \frac{\sum A\bar{y}}{\sum A} & \bar{Y} &= \frac{650\,000}{12\,790} = 50.8 \text{ mm} \end{aligned}$$

CG of a Semicircular Area = $\{0, 4R/3\pi\}$. Try finding it!

ME101 - Division II & IV



19

Center of Mass and Centroids

Theorems of Pappus: Areas and Volumes of Revolution

- method for calculating **surface area** generated by revolving a plane curve about a non-intersecting axis in the plane of the curve
- method for calculating **volume** generated by revolving an area about a non-intersecting axis in the plane of the area

Surface Area

Area of the ring element: circumference times dL
 $dA = 2\pi y dL$

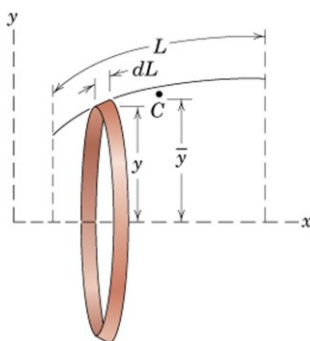
Total area, $A = 2\pi \int y dL$

$\therefore \bar{y}L = \int y dL \rightarrow A = 2\pi \bar{y}L$ or $A = \theta \bar{y}L$ If area is revolved through an angle $\theta < 2\pi$ θ in radians

$\bar{y}$ is the y-coordinate of the centroid C for the line of length L

Generated area is the same as the lateral area of a right circular cylinder of length L and radius $\bar{y}$

Theorems of Pappus can also be used to determine centroid of plane curves if area created by revolving these figures @ a non-intersecting axis is known



ME101 - Division II & IV

Hemant B Kaushik

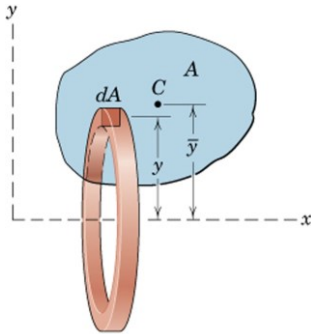
20
L10

Center of Mass and Centroids

Theorems of Pappus: Areas and Volumes of Revolution

- method for calculating **volume** generated by revolving an area about a non-intersecting axis in the plane of the area

Volume



Volume of the ring element: circumference times dA
 $dV = 2\pi y dA$

Total Volume, $V = 2\pi \int y dA$

If area is revolved through an angle $\theta < 2\pi$
 θ in radians

$$\therefore \bar{y}A = \int y dA \rightarrow V = 2\pi \bar{y}A \text{ or } V = \theta \bar{y}A$$

$\bar{y}$ is the y-coordinate of the centroid C of the revolved area A

Generated volume is obtained by multiplying the generating area by the circumference of the circular path described by its centroid.

Theorems of Pappus can also be used to determine centroid of plane areas if volume created by revolving these figures @ a non-intersecting axis is known

ME101 - Division II & IV

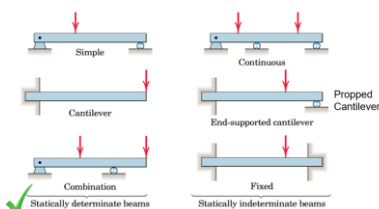
Hemant B Kaushik

21

Beams – Internal Effects

Types of Beams

- Based on support conditions



ME101 - Division II & IV

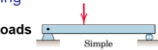
Hemant B Kaushik

55

Types of Beams

- Based on type of external loading

Beams supporting Concentrated Loads



Beams supporting Distributed Loads

- Intensity of distributed load = w

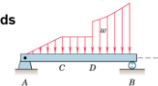
- w is expressed as

force per unit length of beam (N/m)

- intensity of loading may be constant or variable, continuous or discontinuous

- discontinuity in intensity at D (abrupt change)

- At C, intensity is not discontinuous, but rate of change of intensity (dw/dx) is discontinuous



ME101 - Division II & IV

Hemant B Kaushik

56

Beams

Distributed Loads on beams

- General Load Distribution

Differential increment of force is

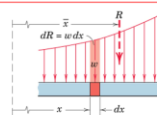
$$dR = w dx$$

Total load R is sum of all the differential forces

$$R = \int w dx \text{ acting at centroid of the area under consideration}$$

$$\bar{x} = \frac{\int xw dx}{R}$$

Once R is known reactions can be found out from Statics



ME101 - Division II & IV

ME101 - Division II & IV

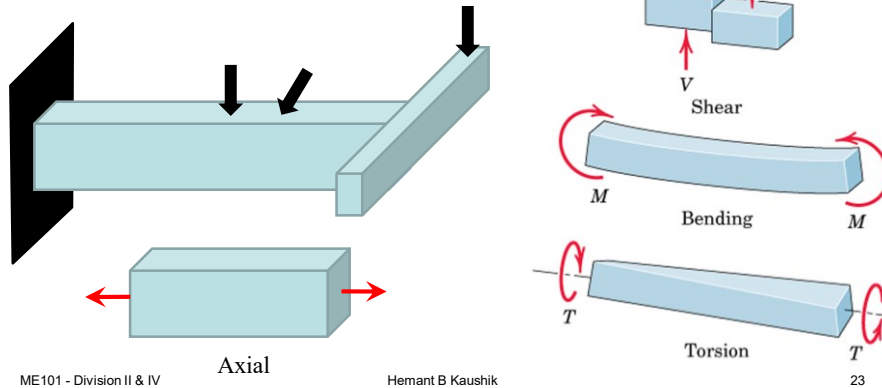
Hemant B Kaushik

58

22

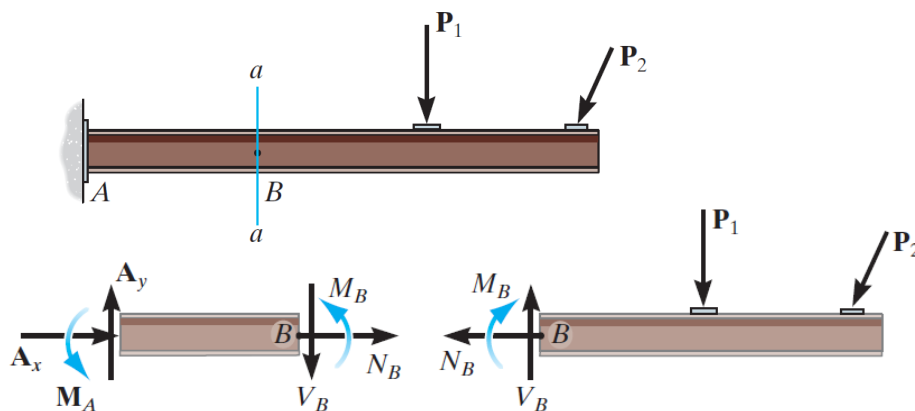
Beams – Internal Effects

- Internal Force Resultants
- Axial Force (N), Shear Force (V), Bending Moment (M), Torsional Moment (T) in Beam
 - Method of Sections is used



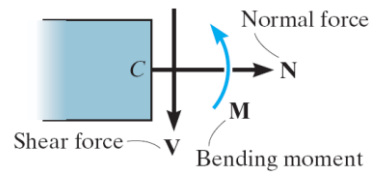
Beams – Internal Effects

- Method of Section:
 - Internal Force Resultants at B → Section a-a at B and use equilibrium equations in both cut parts



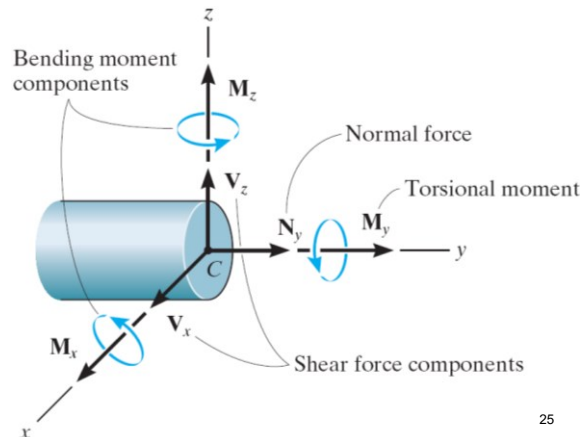
Beams – Internal Effects

- 2D Beam



- 3D Beam

The Force Resultants act at centroid of the section's Cross-sectional area



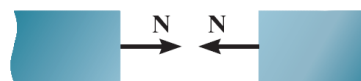
ME101 - Division II & IV

25

Beams – Internal Effects

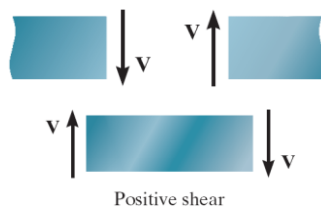
Sign Convention

Positive Axial Force creates Tension



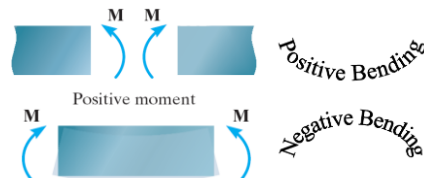
Positive normal force

Positive shear force will cause the Beam segment on which it acts to rotate clockwise



Positive shear

Positive bending moment will tend to bend the segment on which it acts in a concave upward manner (compression on top of section).



Positive moment

Positive Bending

Negative Bending

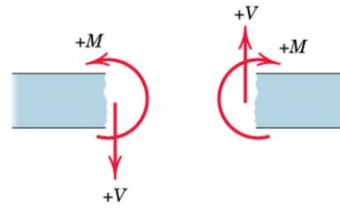
ME101 - Division II & IV

Hemant B Kaushik

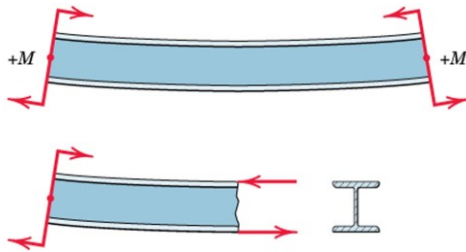
26

Beams – Internal Effects

Sign convention in a single plane



Interpretation of Bending Couple



H-section Beam bent by two equal and opposite positive moments applied at ends

Neglecting resistance offered by web

Compression at top; Tension at bottom

Resultant of these two forces (one tensile and other compressive) acting on any section is a Couple and has the value of the Bending Moment acting on the section.

ME101 - Division II & IV

Hemant B Kaushik

27

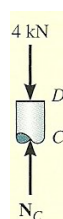
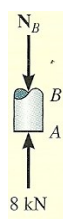
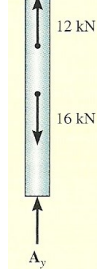
Beams – Internal Effects

Example: Find the axial force in the fixed bar at points B and C

Solution: Draw the FBD of the entire bar

$$+\uparrow \Sigma F_y = 0; \quad A_y - 16 \text{ kN} + 12 \text{ kN} - 4 \text{ kN} = 0 \quad A_y = 8 \text{ kN}$$

Draw sections at B and C to get AF in the bar at B and C

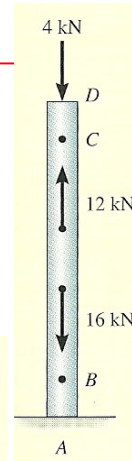


Segment AB

$$+\uparrow \Sigma F_y = 0; \quad 8 \text{ kN} - N_B = 0 \quad N_B = 8 \text{ kN}$$

Segment DC

$$+\uparrow \Sigma F_y = 0; \quad N_C - 4 \text{ kN} = 0 \quad N_C = 4 \text{ kN}$$



Alternatively, take a section at C and consider only CD portion of the bar
Then take a section at B and consider only BD portion of the bar
→ no need to calculate reactions

ME101 - Division II & IV

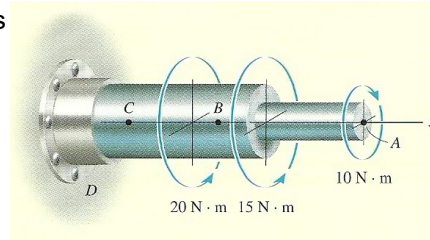
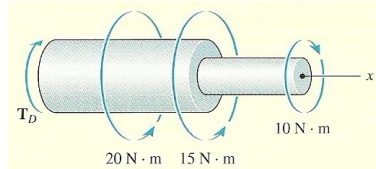
Hemant B Kaushik

28

Beams – Internal Effects

Example: Find the internal torques at points B and C of the circular shaft subjected to three concentrated torques

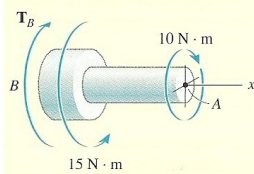
Solution: FBD of entire shaft



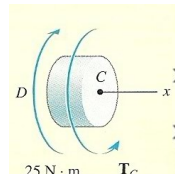
$$\Sigma M_x = 0; \quad -10 \text{ N} \cdot \text{m} + 15 \text{ N} \cdot \text{m} + 20 \text{ N} \cdot \text{m} - T_D = 0$$

$$T_D = 25 \text{ N} \cdot \text{m}$$

Sections at B and C and FBDs of shaft segments AB and CD



ME101 - Division II & IV



Hemant B Kaushik

Segment AB

$$\Sigma M_x = 0; \quad -10 \text{ N} \cdot \text{m} + 15 \text{ N} \cdot \text{m} - T_B = 0 \quad T_B = 5 \text{ N} \cdot \text{m}$$

Segment CD

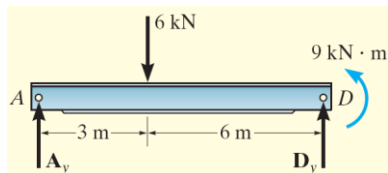
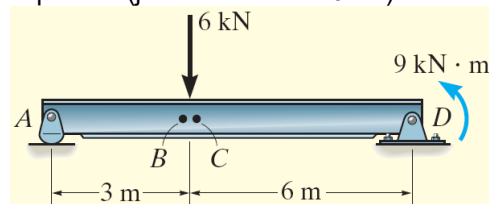
$$\Sigma M_x = 0; \quad T_C - 25 \text{ N} \cdot \text{m} = 0 \quad T_C = 25 \text{ N} \cdot \text{m}$$

29

Beams – Internal Effects

Example: Find the AF, SF, and BM at point B (just to the left of 6 kN) and at point C (just to the right of 6 kN)

Solution: Draw FBD of entire beam



$$\zeta + \Sigma M_D = 0; \quad 9 \text{ kN} \cdot \text{m} + (6 \text{ kN})(6 \text{ m}) - A_y(9 \text{ m}) = 0$$

$$A_y = 5 \text{ kN}$$

D_y need not be determined if only left part of the beam is analysed

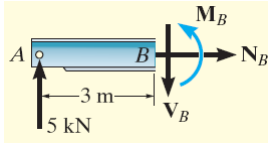
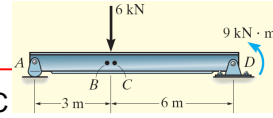
ME101 - Division II & IV

Hemant B Kaushik

30

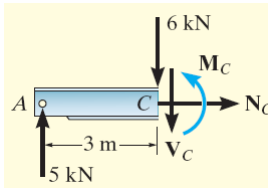
Beams – Internal Effects

Example Solution: Draw FBD of segments AB and AC and use equilibrium equations



Segment AB

$$\begin{aligned} \rightarrow \sum F_x &= 0; & N_B &= 0 \\ +\uparrow \sum F_y &= 0; & 5 \text{ kN} - V_B &= 0 & V_B &= 5 \text{ kN} \\ \curvearrowleft \sum M_B &= 0; & -(5 \text{ kN})(3 \text{ m}) + M_B &= 0 & M_B &= 15 \text{ kN} \cdot \text{m} \end{aligned}$$



Segment AC

$$\begin{aligned} \rightarrow \sum F_x &= 0; & N_C &= 0 \\ +\uparrow \sum F_y &= 0; & 5 \text{ kN} - 6 \text{ kN} - V_C &= 0 & V_C &= -1 \text{ kN} \\ \curvearrowleft \sum M_C &= 0; & -(5 \text{ kN})(3 \text{ m}) + M_C &= 0 & M_C &= 15 \text{ kN} \cdot \text{m} \end{aligned}$$

ME101 - Division II & IV

Hemant B Kaushik

31

Beams – SFD and BMD

Shear Force Diagram and bending Moment Diagram

- Variation of SF and BM over the length of the beam
→ SFD and BMD
- Maximum magnitude of BM and SF and their locations is prime consideration in beam design
- SFDs and BMDs are plotted using method of section
 - Equilibrium of FBD of entire Beam
→ External Reactions
 - Equilibrium of a cut part of beam
→ Expressions for SF and BM at the cut section

Use the positive sign convention consistently

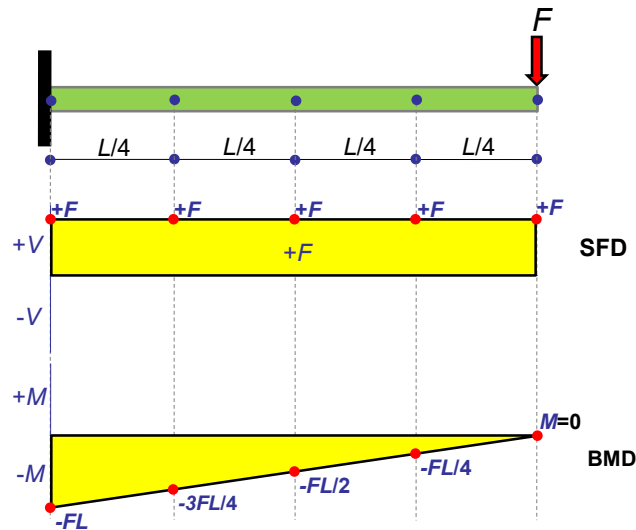
ME101 - Division II & IV

Hemant B Kaushik

32
L11

Beams – SFD and BMD

Draw SFD and BMD for a cantilever beam supporting a point load at the free end



ME101 - Division II & IV

Hemant B Kaushik

33

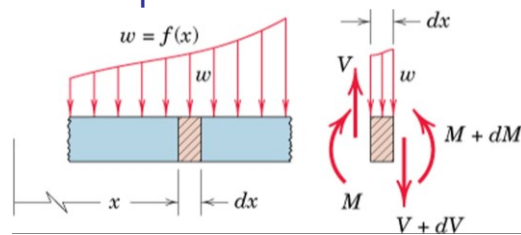
Beams – SFD and BMD

Shear and Moment Relationships

Consider a portion of a beam

Isolate an element dx

Draw FBD of the element



Vertical equilibrium in $dx \rightarrow$

$$V - w dx - (V + dV) = 0 \quad w = -\frac{dV}{dx}$$

Moment equilibrium in dx ($\sum M @$ the left side of the element) $\rightarrow$

$$M + w dx \frac{dx}{2} + (V + dV) dx - (M + dM) = 0 \quad V = \frac{dM}{dx}$$

Terms $w(dx)^2/2$ and $dV dx$ may be dropped since they are differentials of higher order than those which remains.

ME101 - Division II & IV

Hemant B Kaushik

34

Beams – SFD and BMD

Shear and Moment Relationships

$$w = -\frac{dV}{dx}$$

Slope of the shear diagram = - Value of applied loading

$$V = \frac{dM}{dx}$$

Slope of the moment curve = Shear Force

Both equations not applicable at the point of loading because of discontinuity produced by the abrupt change in shear.

ME101 - Division II & IV

Hemant B Kaushik

35

Beams – SFD and BMD

Shear and Moment Relationships

Expressing V in terms of w by integrating $w = -\frac{dV}{dx}$

$$\int_{V_0}^V dV = -\int_{x_0}^x w dx \quad \text{OR} \quad V = V_0 + (\text{the negative of the area under the loading curve from } x_0 \text{ to } x)$$

V_0 is the shear force at x_0 and V is the shear force at x

Expressing M in terms of V by integrating $V = \frac{dM}{dx}$

$$\int_{M_0}^M dM = \int_{x_0}^x V dx \quad \text{OR} \quad M = M_0 + (\text{area under the shear diagram from } x_0 \text{ to } x)$$

M_0 is the BM at x_0 and M is the BM at x

ME101 - Division II & IV

Hemant B Kaushik

36

Beams – SFD and BMD

$$V = V_0 + (\text{negative of area under the loading curve from } x_0 \text{ to } x)$$

$$M = M_0 + (\text{area under the shear diagram from } x_0 \text{ to } x)$$

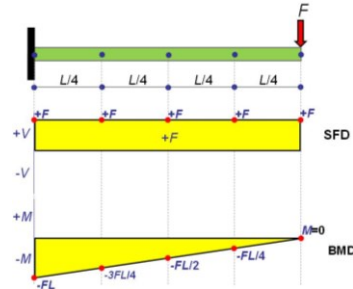
If there is no externally applied moment M_0 at $x_0 = 0$, total moment at any section equals the area under the shear diagram up to that section

When V passes through zero and is a continuous function of x with $dV/dx \neq 0$ (i.e., nonzero loading)

$$\rightarrow \frac{dM}{dx} = 0$$

$\rightarrow$ BM will be a maximum or minimum at this point

Critical values of BM also occur when SF crosses the zero axis discontinuously (Example: Beams under concentrated loads)



ME101 - Division II & IV

Hemant B Kaushik

37

Beams – SFD and BMD

$$w = -\frac{dV}{dx} \quad \text{Degree of } V \text{ in } x \text{ is one higher than that of } w$$

$$V = \frac{dM}{dx} \quad \text{Degree of } M \text{ in } x \text{ is one higher than that of } V$$

$\rightarrow$ Degree of M in x is two higher than that of w

Combining the two equations $\rightarrow \frac{d^2 M}{dx^2} = -w$

$\rightarrow$ If w is a known function of x , BM can be obtained by integrating this equation twice with proper limits of integration.

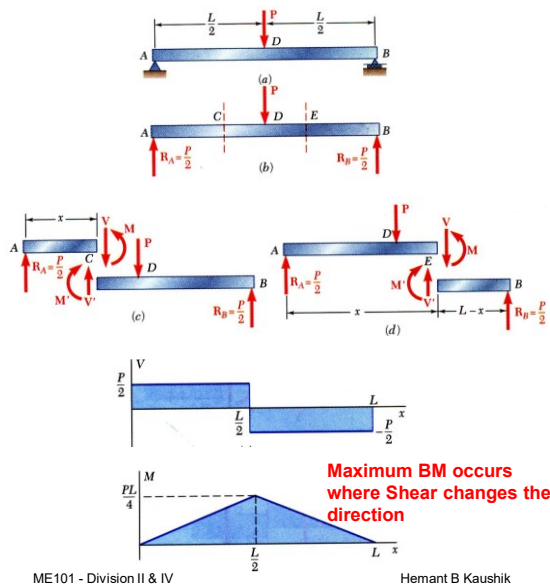
$\rightarrow$ Method is usable only if w is a continuous function of x (other cases not part of this course)

ME101 - Division II & IV

Hemant B Kaushik

38

Beams – SFD and BMD: Example



- Draw the SFD and BMD.
- Determine reactions at supports.
- Cut beam at C and consider member AC ,
 $V = +P/2$ $M = +Px/2$
- Cut beam at E and consider member EB ,
 $V = -P/2$ $M = +P(L-x)/2$
- For a beam subjected to concentrated loads, shear is constant between loading points and moment varies linearly.

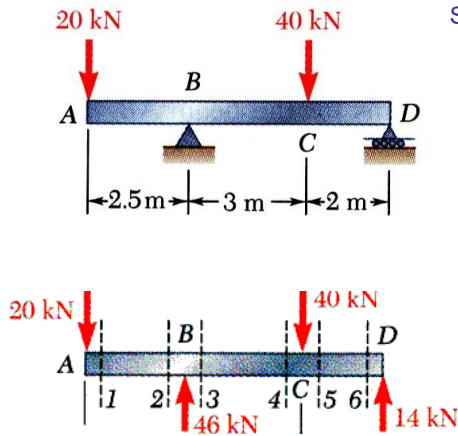
ME101 - Division II & IV

Hemant B Kaushik

39

Beams – SFD and BMD: Example

Draw the shear and bending moment diagrams for the beam and loading shown.



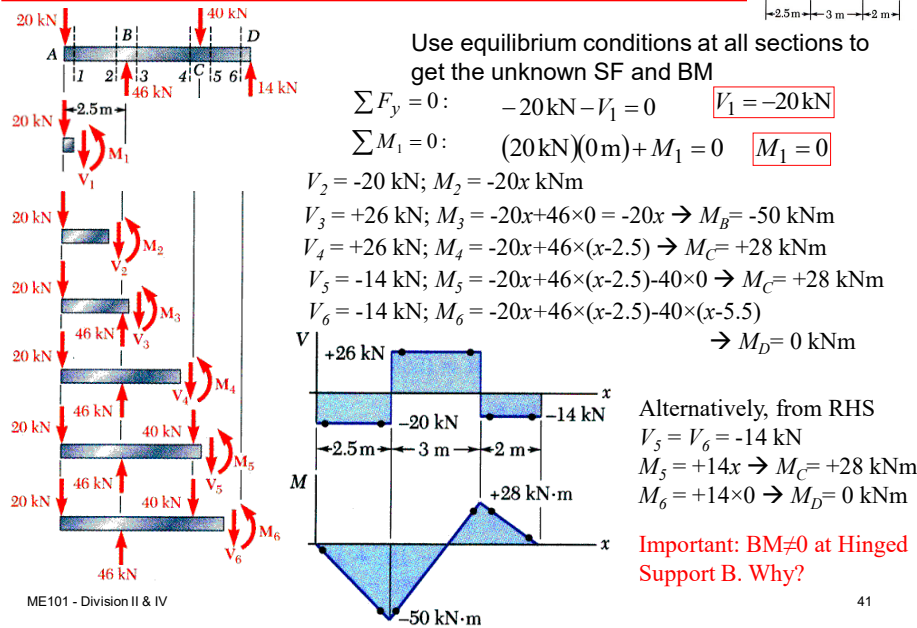
Solution: Draw FBD and find out the support reactions using equilibrium equations

ME101 - Division II & IV

Hemant B Kaushik

40
L12

Beams – SFD and BMD: Example

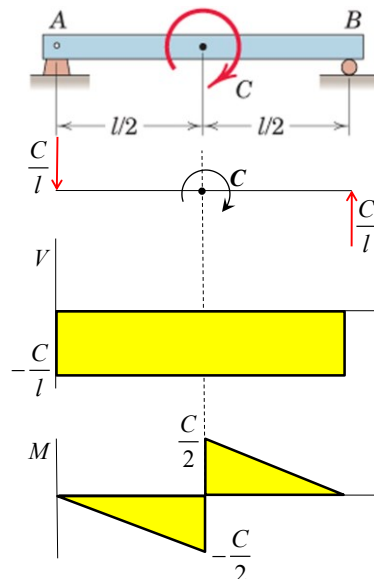


Beams – SFD and BMD: Example

Draw the SFD and BMD for the beam acted upon by a clockwise couple at mid point

Solution: Draw FBD of the beam and Calculate the support reactions

Draw the SFD and the BMD starting From any one end



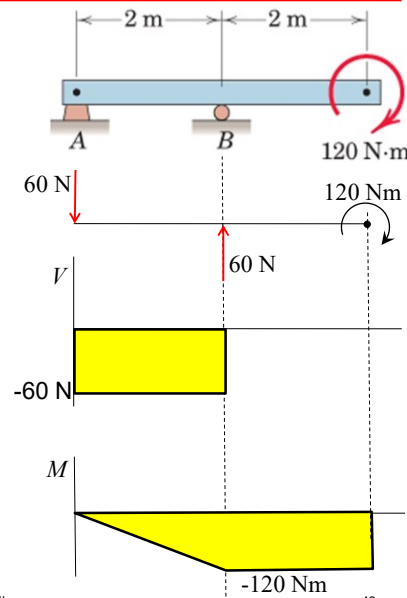
Beams – SFD and BMD: Example

Draw the SFD and BMD for the beam

Solution: Draw FBD of the beam and
Calculate the support reactions

$$\begin{aligned}\sum M_A = 0 &\rightarrow R_B = 60 \text{ N} \uparrow \\ \sum M_B = 0 &\rightarrow R_A = 60 \text{ N} \downarrow\end{aligned}$$

Draw the SFD and the BMD starting
from any one end



ME101 - Division II & IV

Hemant B Kaushik

43

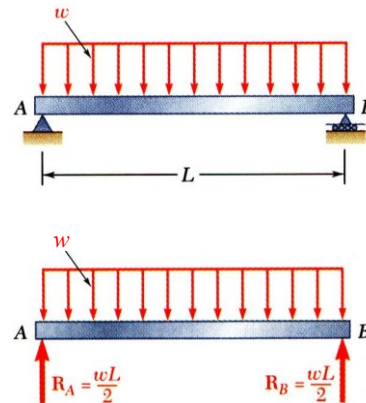
Beams – SFD and BMD: Example

Draw the SFD and BMD for the beam

Solution:

Draw FBD of the entire beam
and calculate support reactions
using equilibrium equations

$$\text{Reactions at supports: } R_A = R_B = \frac{wL}{2}$$



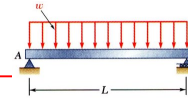
Develop the relations between loading, shear force, and bending
moment and plot the SFD and BMD

ME101 - Division II & IV

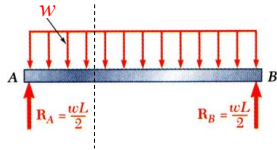
Hemant B Kaushik

44

Beams – SFD and BMD: Example



Shear Force at any section: $V = \frac{wL}{2} - wx = w\left(\frac{L}{2} - x\right)$



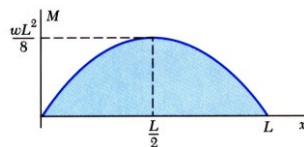
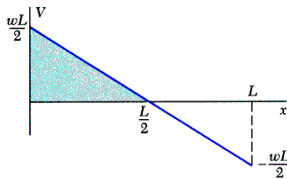
Alternatively, $V - V_A = -\int_0^x w dx = -wx$

$$V = V_A - wx = \frac{wL}{2} - wx = w\left(\frac{L}{2} - x\right)$$

BM at any section: $M = \frac{wL}{2}x - wx\frac{x}{2} = \frac{w}{2}(Lx - x^2)$

Alternatively, $M - M_A = \int_0^x V dx$

$$M = \int_0^x w\left(\frac{L}{2} - x\right) dx = \frac{w}{2}(Lx - x^2)$$



$$M_{\max} = \frac{wL^2}{8} \quad \because \left(M \text{ at } \frac{dM}{dx} = V = 0 \right)$$

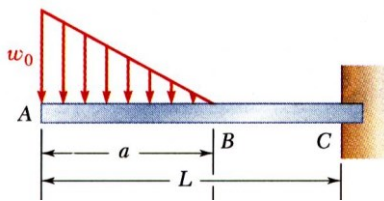
ME101 - Division II & IV

Hemant B Kaushik

45

Beams – SFD and BMD: Example

Draw the SFD and BMD for the Beam

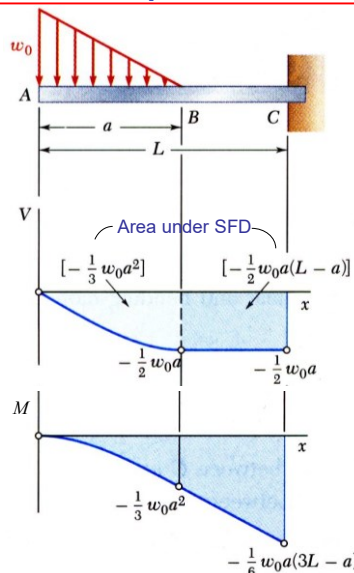


Solution:

SFD and BMD can be plotted without determining support reactions since it is a cantilever beam.

However, values of SF and BM can be verified at the support if support reactions are known.

$$R_C = \frac{w_0 a}{2} \uparrow; \quad M_C = \frac{w_0 a}{2} \left(L - \frac{a}{3} \right) = \frac{w_0 a}{6} (3L - a) \downarrow$$



ME101 - Division II & IV

Hemant B Kaushik

46

